



Six-Junction Solar Cells Can Reach Astounding Efficiency Of 47 Per Cent

A solar cell is a sandwich of two different layers of silicon that have been specially treated or doped to let electricity flow through them in a particular way. This article covers the working principle, architecture, and uses of advanced multi-junction solar cells in achieving efficiency like never before



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Single-junction flat-plate terrestrial solar cells are fundamentally limited to about 25 per cent solar to electricity conversion efficiency. Scientists at the National Renewable Energy Laboratory (NREL), USA, have fabricated a solar cell that now holds the world record for the highest solar conversion efficiency of 47.1 per cent using a monolithic, series-connected, six-junction inverted metamorphic structure operated under the direct spectrum at 143 sun concentration. A variation of this structure achieves a one-sun global efficiency of 39.2 per cent. One sun is equivalent to $100\text{mW}/\text{cm}^2$ of irradiance.

Working principle of solar cell

A solar cell is a sandwich of two different layers of silicon that have been specially treated or doped to let electricity flow through them in a particular way (Fig. 1).

The lower layer is doped in such a way that it has very few electrons. It is called p-type or positive type. The upper layer is doped the opposite way to give it lots of electrons. It is called n-type or negative type. A barrier is created at the junction of the two layers, and

no electrons can cross this barrier.

But something remarkable happens when the sun shines on the sandwich. Streams of energetic 'light particles' called photons enter the sandwich. They give up their energy to the atoms of semiconductors, which knocks electrons out of the lower p-type layer.

Usually, these electrons are in the valence band. The energy given to electrons by photons excites them into the conduction band, leaving behind an empty space called 'hole.' An electron from a neighbouring atom can move into this hole. The dislocated electrons jump across the barrier to the n-type layer above and flow out around the circuit to maintain the equilibrium, as shown in Fig. 2.

Band gap

The lower energy level of a semiconductor is called the 'valence band,' and the energy level at which an electron is considered free is called the 'conduction band.' A band gap is the energy difference in electron volts between the top of the valence band and the bottom of the conduction band. It represents the minimum energy that is required to excite an electron up to a state in the conduction band where it can participate in conduction.

Semiconductor materials are nominally small band gap insulator. These materials can be doped with impurities that alter their electronic properties in a controllable way. An electric volt is the amount of kinetic energy gained or lost by a single electron accelerating from rest through an

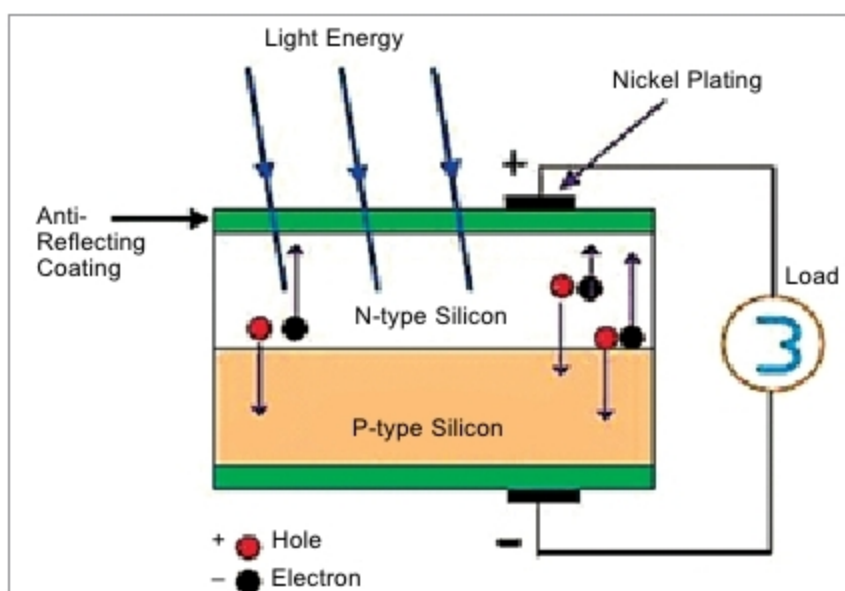


Fig. 1: Schematic diagram of a solar cell